

1 Introduction

Handwritten digit recognition is a fundamental problem in the field of pattern recognition and computer vision. It has practical applications in areas such as document digitization, automatic number recognition, and financial systems.

In this project, a deep learning-based approach is employed to classify Bengali handwritten digits (0–9). The goal is to develop a robust model capable of accurately recognizing digit patterns from grayscale images. The model performance is evaluated using the macro F1-score, which ensures balanced evaluation across all classes.

2 Dataset Description

2.1 Dataset Summary

Attribute	Value
Total Samples	36,682
Number of Classes	10 (0–9)
Image Type	Grayscale PNG
Data Source	CSV + Image Folder

The dataset consists of grayscale images representing Bengali handwritten digits. Each image is associated with a label indicating the correct digit. The dataset is organized using CSV files that map image filenames to their corresponding labels.

3 Methodology

3.1 Data Preprocessing

To ensure consistent input and improve model performance, several preprocessing techniques were applied:

- Histogram equalization to enhance image contrast and highlight digit features
- Resizing all images to 64×64 pixels for uniformity
- Normalization of pixel values to stabilize training
- Data augmentation techniques such as rotation and translation to increase dataset diversity

These preprocessing steps reduce noise, improve feature visibility, and help the model generalize better to unseen data.

3.2 Model Architecture

Layer	Configuration	Purpose
Conv1	32 filters, 3×3	Extract low-level features
Conv2	64 filters, 3×3	Capture intermediate patterns
Conv3	128 filters, 3×3	Learn complex features
MaxPooling	2×2	Reduce spatial dimensions
Fully Connected	256 neurons	Combine learned features
Dropout	0.3	Prevent overfitting
Output Layer	10 neurons	Perform classification

The custom Convolutional Neural Network (CNN) extracts hierarchical features from the input images. Convolutional layers detect edges and patterns, while fully connected layers map these features to output classes.

3.3 Training Configuration

Parameter	Value
Optimizer	Adam
Loss Function	CrossEntropy Loss
Learning Rate	1×10^{-3}
Batch Size	32
Epochs	15

The Adam optimizer was chosen for its efficiency in handling large datasets. CrossEntropy loss is suitable for multi-class classification tasks.

4 Evaluation Metrics

The macro F1-score is used as the primary evaluation metric:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score balances precision and recall and is particularly useful when evaluating classification performance across multiple classes.

5 Results

5.1 Training Performance

Epoch	Loss	F1-score
1	1.0878	0.8969
2	0.4577	0.9370
3	0.3355	0.9482
4	0.2771	0.9533
5	0.2380	0.9665
6	0.2200	0.9646
7	0.1981	0.9664
8	0.1823	0.9697
9	0.1637	0.9741
10	0.1576	0.9733
11	0.1530	0.9741
12	0.1485	0.9708
13	0.1417	0.9774
14	0.1366	0.9742
15	0.1314	0.9735

The results show a steady decrease in training loss and a consistent improvement in validation F1-score, demonstrating effective learning and good convergence of the model. The best validation macro F1-score achieved is 0.9774 at epoch 13.

5.2 Classification Report

The classification report shows high precision, recall, and F1-score values for all digit classes. This indicates that the model performs consistently well across different classes without significant bias.

Digit	Precision	Recall	F1-score	Support
0	0.98	0.98	0.98	368
1	0.94	0.96	0.95	365
2	0.99	0.99	0.99	367
3	0.98	0.98	0.98	368
4	0.98	0.97	0.98	369
5	0.97	0.98	0.97	368
6	0.98	0.97	0.98	369
7	0.97	0.99	0.98	366
8	1.00	0.99	1.00	364
9	0.97	0.95	0.96	365
Overall				
Accuracy	0.98			
Macro Avg	0.98	0.98	0.98	3669
Weighted Avg	0.98	0.98	0.98	3669

5.3 Confusion Matrix

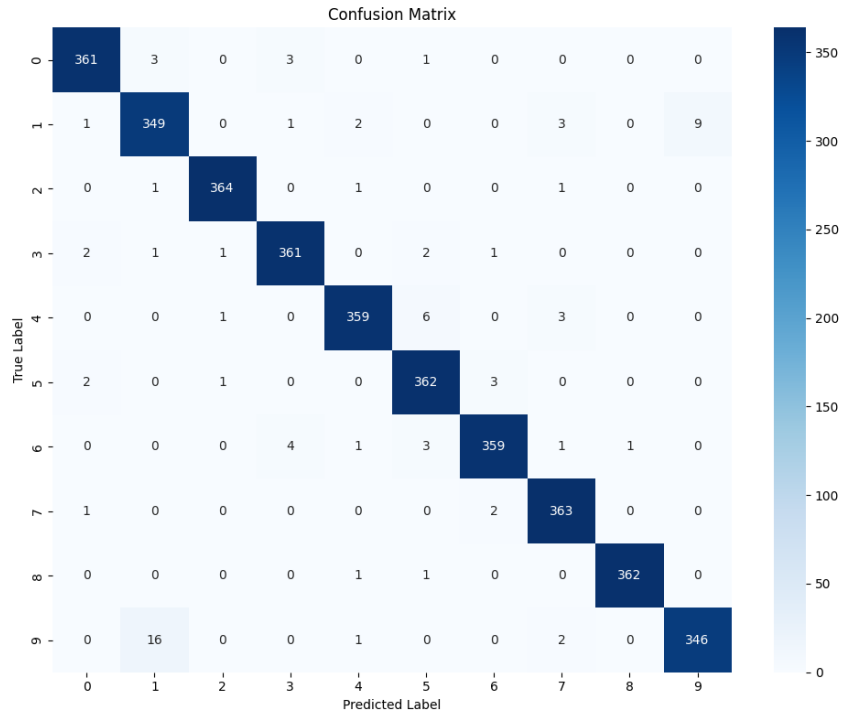


Figure 1: Confusion Matrix of the Proposed Model

The confusion matrix shows that most predictions are concentrated along the diagonal, indicating correct classification. Only minor misclassifications are observed between visually similar digits.

6 Conclusion

The custom CNN model developed in this competition achieved a strong macro F1-score of **0.9774** on the validation set, demonstrating excellent classification performance for Bengali handwritten digits. The combination of histogram equalization for contrast enhancement, data augmentation (rotation and translation), and the three-layer convolutional architecture played a vital role in improving feature extraction and model generalization. The network successfully captured both low-level edges and higher-level digit patterns, resulting in consistently high precision, recall, and F1-scores across all classes.

Overall, the proposed model proved highly effective for the task of Bengali handwritten digit recognition. Future work may focus on exploring deeper architectures such as ResNet, applying transfer learning from pre-trained models, implementing ensemble techniques, or incorporating more advanced augmentation strategies to further enhance performance and robustness.